

# ISHAAN BALANI

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## EDUCATION

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**Bachelor of Computer Science**, Arizona State University

Expected 2027

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Software Engineering, Cybersecurity

## EXPERIENCE

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**Undergraduate Research Assistant**

August 2024 - Present

*Tempe, AZ*

- Optimized AI video generation pipeline, reducing processing time by 30% through scalable Tensorflow and Pytorch implementations.
- Applied 3D computer vision (OpenCV) on datasets of over 50,000 frames, enabling automated feature extraction and improving synthesis accuracy by 15%.
- Authored modular research code that was reused in 2 other research lab projects ensuring maintainability and reproducibility.

**Computer Science Team Lead Intern**

October 2024 - July 2025

*Remote/Tempe, AZ*

- Directed a team of 5 developers to build a MERN-stack platform serving over 1,200 high school users for scholarship discovery.
- Increased backend query efficiency by 40% via database indexing and optimized API calls.
- Reduced average bug resolution time from 3-5 days to <24 hours by implementing CI/CD workflows and automated testing.

## PROJECTS ([GITHUB](#))

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**TaskLocker**

*Flask, Javascript, Google Gemini API*

- Designed a scheduling assistant used by over 100 ASU students, parsing PDF schedules into Google Calendar.
- Cut routine planning time by 90% with automated Google Gemini integration.
- Built secure Flask APIs supporting over 500 successful schedulign requests with <1% error rate.

**AI Tumor Identifier**

*Python, Scikit-learn*

- Built an ML pipeline achieving over 99% accuracy across over 5,000 tumor records using decision trees and SVM classifiers.
- Automated preprocessing to cut manual data cleaning effort by 50%, improving reproducibility.

**Music Recommendation Model (UC Berkeley Hackathon)**

*Python, Flask, Pytorch*

- Led a team of 4 to build a real-time music recommendation engine analyzing over 10,000 tracks.
- Deployed a Flask-based demo app with a <200ms response time for song recommendations during live judging.

## CERTIFICATIONS + SKILLS

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**Certifications:**

- CCST: Cisco Certified Support Technician (Networking)

- Cisco Networking Academy: Introduction to Cybersecurity
- Cisco Networking Academy: Introduction to Modern AI

#### **Technical Skills:**

- **Languages/Frameworks:** Python, C, C++, Java, Flask, React, JavaScript, SQL, TensorFlow, PyTorch, OpenCV
- **Systems & Tools:** Linux, Git, AWS (EC2/S3), Docker, MongoDB, REST APIs, Google Cloud
- **Core Knowledge:** Operating Systems, Networking Fundamentals (TCP/IP, DNS, DHCP), Distributed Systems, Information Security
- **Soft Skills:** Leadership, Collaboration, Communication, Problem-Solving